

Group A Practical 1

```
import java.util.*;

public class Conflation {

    private static List<String> textList = new ArrayList<>();

    private static Scanner scanner = new Scanner(System.in);

    public static void main(String[] args) {

        // Initialize with sample text

        textList = Arrays.asList("she", "went", "to", "the", "store", "and", "bought", "fruits");

        int choice;

        do {

            System.out.println("\nText Processing Menu:");

            System.out.println("1. Display text");

            System.out.println("2. Remove punctuation and stop words");

            System.out.println("3. Suffix stripping");

            System.out.println("4. Count word frequency");

            System.out.println("5. Exit");

            System.out.print("Enter your choice: ");

            try {

                choice = scanner.nextInt();

                scanner.nextLine(); // Consume newline
```

```
switch (choice) {  
    case 1:  
        displayText();  
        break;  
    case 2:  
        removePunctuationAndStopWords();  
        break;  
    case 3:  
        suffixStripping();  
        break;  
    case 4:  
        countFrequency();  
        break;  
    case 5:  
        System.out.println("Exiting program...");  
        break;  
    default:  
        System.out.println("Invalid choice. Please try again.");  
}  
} catch (InputMismatchException e) {  
    System.out.println("Invalid input. Please enter a number.");  
    scanner.nextLine(); // Clear the invalid input  
    choice = 0; // Continue the loop  
}  
} while (choice != 5);
```

```
    scanner.close();  
}
```

```
private static void displayText() {  
    System.out.println("Current text:");  
    for (String word : textList) {  
        System.out.print(word + " ");  
    }  
    System.out.println();  
}
```

```
private static void removePunctuationAndStopWords() {  
    // Removing punctuation and making the word lowercase  
    List<String> stopWords = Arrays.asList("a", "an", "the", "and", "or", "but");  
    List<String> processedText = new ArrayList<>();  
  
    for (String word : textList) {  
        // Remove punctuation  
        String cleanWord = word.replaceAll("[^a-zA-Z]", "").toLowerCase();  
  
        // Check if it's not a stop word  
        if (!stopWords.contains(cleanWord) && !cleanWord.isEmpty()) {  
            processedText.add(cleanWord);  
        }  
    }  
}
```

```
textList = processedText;

System.out.println("Punctuation and stop words removed.");

displayText();
}
```

```
private static void suffixStripping() {

    // Strip suffixes

    List<String> processedText = new ArrayList<>();

    for (String word : textList) {

        String strippedWord = word;

        // Common suffixes to strip

        String[] suffixes = {"ing", "ed", "er", "est", "ly", "tion", "ness"};

        for (String suffix : suffixes) {

            if (strippedWord.endsWith(suffix) && strippedWord.length() > suffix.length()) {

                strippedWord = strippedWord.substring(0, strippedWord.length() -
suffix.length());

                break; // Only strip one suffix

            }

        }

        processedText.add(strippedWord);

    }

}
```

```

    textList = processedText;

    System.out.println("Suffixes stripped.");
    displayText();
}

private static void countFrequency() {
    // Count word frequency

    Map<String, Integer> frequencyMap = new HashMap<>();

    for (String word : textList) {
        frequencyMap.put(word, frequencyMap.getOrDefault(word, 0) + 1);
    }

    System.out.println("Word frequency:");
    for (Map.Entry<String, Integer> entry : frequencyMap.entrySet()) {
        System.out.println(entry.getKey() + ": " + entry.getValue());
    }
}
}

```

Output

Text Processing Menu:

1. Display text
2. Remove punctuation and stop words

3. Suffix stripping

4. Count word frequency

5. Exit

Enter your choice: 1

Current text:

she went to the store and bought fruits

Text Processing Menu:

1. Display text

2. Remove punctuation and stop words

3. Suffix stripping

4. Count word frequency

5. Exit

Enter your choice: 2

Punctuation and stop words removed.

Current text:

she went to store bought fruits

Text Processing Menu:

1. Display text

2. Remove punctuation and stop words

3. Suffix stripping

4. Count word frequency

5. Exit

Enter your choice: